

Chapter 12: Ratios, rates and time

Drill practice and mixed revision of all 11 lessons · 98
questions

NSW syllabus outcome

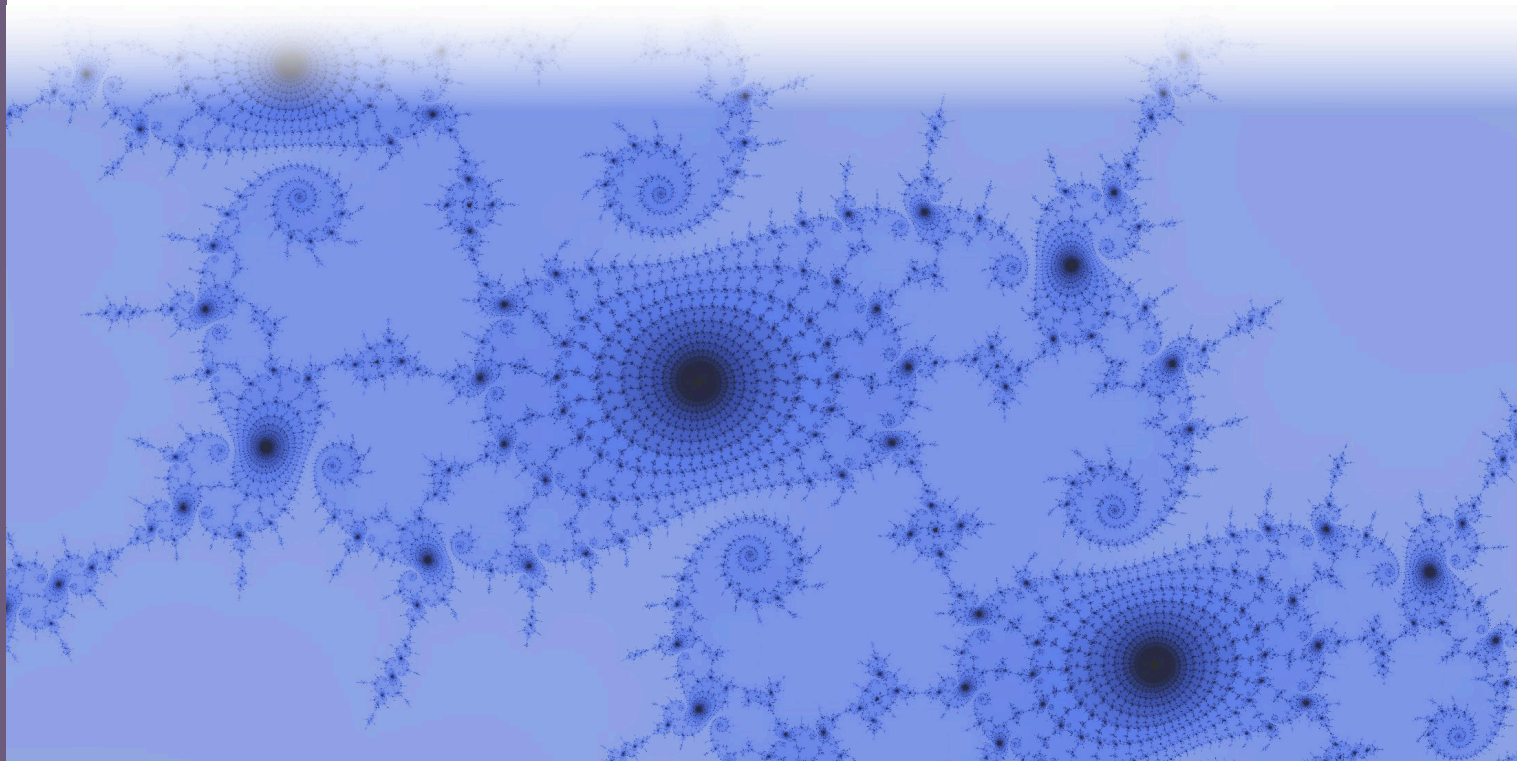
Stage 4 Number and algebra: MA4-RAT-C-01 solves problems
involving ratios and rates, and analyses distance–time graphs;
time calculations from Stage 3 extended.

NAME

CLASS

CHAPTER

12



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 12. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–31

12.01 Ratios**Easy** Write the ratio (do not simplify).**1** 7 green, 4 blue: green to blue
_____**2** 2 red, 12 blue: red to blue
_____**3** 2 green, 14 red: red to green
_____**Medium** Change to the same unit, then write the ratio in simplest form.**4** 100 g to 3 kg
_____**5** 160 min to 2 h
_____**6** 10 mm to 5 cm
_____**12.02 Equivalent and simplified ratios****Easy** Simplify the ratio.**7** 12 : 32
_____**8** 33 : 36
_____**9** 42 : 72
_____**10** 108 : 63
_____**Medium** Simplify. Change decimals or fractions to whole numbers first.**11** 0.1 : 0.2
_____**12** 0.6 : 0.1
_____**13** 0.4 : 0.2
_____**14** 0.7 : 0.8
_____**12.03 Ratio problems****Easy** Share the amount in the ratio given.**15** \$48 in 3 : 3
_____**16** \$60 in 4 : 2
_____**17** \$36 in 4 : 2
_____**Medium** Share the quantity in the ratio given.**18** 20 m in 5 : 4 : 1
_____**19** 30 m in 5 : 2 : 3
_____**20** 40 kg in 2 : 5 : 3
_____**12.04 Rates****Easy** Write as a rate for one unit.**21** \$80 in 5 kg
_____**22** 102 m in 3 s
_____**23** 200 m in 8 s
_____**Medium** Convert the rate.**24** 90 km/h to km/min
_____**25** \$30/kg to c/g
_____**26** \$18/kg to c/g
_____**12.05 Best buys****Easy** Find the unit price (the cost of one).**27** 4 for \$25.32
_____**28** 8 for \$18.24
_____**29** 4 for \$33.60
_____**Medium** Which is the better buy? Find the price per 100 g (or per 100 mL).**30** 300 g for \$2.12 or 1000 g for \$10.69
_____**31** 250 g for \$1.88 or 500 g for \$5.98

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 32–63

12.06 Rate problems

Easy Use the rate to find the total.

32 3 h at \$9/h

33 3 kg at \$13/kg

34 12 h at 115 km/h

Medium Find the distance: distance = speed × time.

35 120 km/h for 30 min

36 90 km/h for 2 h 30 min

37 80 km/h for 30 min

12.07 Travel graphs

Easy Find the speed: speed = distance ÷ time.

38 60 km in 3 h

39 600 km in 6 h

40 500 km in 5 h

Medium Find the missing value: $d = s \times t$, $s = d \div t$, $t = d \div s$.

41 $d = 80$ km, $t = 2$ h, $s = ?$

42 $s = 40$ km/h, $t = 2$ h, $d = ?$

43 $d = 80$ km, $s = 20$ km/h, $t = ?$

12.08 Time

Easy Convert. 1 h = 60 min, 1 min = 60 s.

44 6 h to min

45 490 min to h and min

46 430 min to h and min

Medium Subtract the times.

47 3 h 40 min – 1 h 49 min

48 2 h 19 min – 29 min

49 4 h 1 min – 1 h 57 min

12.09 12-hour and 24-hour time

Easy Write in 24-hour time.

50 6:00 a.m.

51 7:25 p.m.

52 4:55 a.m.

53 6:35 p.m.

Medium Convert. Take care near midnight and midday.

54 11:55 p.m.

55 11:25 a.m.

56 12:00 a.m.

57 23:30

12.10 Time differences

Easy Find the time difference.

58 8:30 a.m. to 3:35 p.m.

59 8:10 a.m. to 12:35 p.m.

60 12:50 p.m. to 7:00 p.m.

Medium Find the time difference. It goes past midnight.

61 21:40 to 01:45

62 23:55 to 03:50

63 21:00 to 01:35

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 64–70

12.11 Timetables

Easy How long is each trip?

64 13:35 → 14:40

65 08:40 → 11:10

66 16:00 → 16:55

67 13:30 → 14:20

Medium How long is each trip? It goes past the hour.

68 4:40 p.m. → 5:25 p.m.

69 4:45 p.m. → 6:30 p.m.

70 10:50 a.m. → 12:45 p.m.

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 71–80

71

Easy

A bag has 8 red and 5 blue marbles. Write the ratio.
all to red Lesson 12.01

72

Easy

Convert.
6:25 p.m. to 24-hour Lesson 12.09

73

Easy

Find how many or how long.
1000 L at 40 L/min Lesson 12.06

74

Easy

Share \$60 in the ratio.
1 : 4 Lesson 12.03

75

Easy

How long is the trip?
10:35 → 12:10 Lesson 12.11

76

Easy

Convert.
200 min to h and min Lesson 12.08

77

Easy

Find the unit price.
8 for \$10 Lesson 12.05

78

Easy

Simplify the ratio.
16 : 24 Lesson 12.02

79

Easy

Find the time difference.
10:40 a.m. to 3:15 p.m. Lesson 12.10

80

Easy

Find the speed.
90 km in 2 h Lesson 12.07

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 81–90

81 **Easy**

Write as a rate per 1 unit.

500 m in 100 s Lesson 12.04

82 **Medium**

Write each ratio.

A 9 m pole and a 4 m pole: short to long Lesson 12.01

83

Medium Convert 8:05 p.m. to 24-hour time.

Lesson 12.09

84

Medium How long to travel 180 km at 60 km/h?

Lesson 12.06

85

Medium Share \$80 in the ratio 3 : 5. Lesson 12.03

86

Medium

Use the bus timetable.

Write the times for the 16:10 bus in 12-hour time.

Lesson 12.11

Stop	Bus times (24-hour)		
School	15:10	15:40	16:10
Library	15:18	15:48	16:18
Shops	15:31	16:01	16:31
Beach	15:45	16:15	16:45

87

Medium Evaluate 2 h 40 min + 1 h 35 min.

Lesson 12.08

88

Medium Which is better: 1 L for \$3 or 2 L for \$5.40?

Lesson 12.05

89

Medium Simplify 16 : 40. Lesson 12.02

90

Medium Find the time from 22:30 to 01:10.

Lesson 12.10

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 91–98

- 91** **Medium** A graph goes from 0 km at 1 p.m. to 30 km at 3 p.m. Find the speed. Lesson 12.07

- 93** **Challenging** A juice is 1 part concentrate to 6 parts water. Write the ratio of water to the whole drink. Lesson 12.01

- 95** **Challenging** Which is better: 250 g for \$2.50 or 1 kg for \$9? Lesson 12.05

- 97** **Challenging** Convert 12:45 a.m. to 24-hour time. Lesson 12.09

- 92** **Medium** A car travels 150 km in 2 h. Find its speed. Lesson 12.04

- 94** **Challenging** Juice : water = 1 : 4. How much water for 250 mL of juice? Lesson 12.03

- 96** **Challenging**
Think it through.
What does a line going down to the time axis show? Lesson 12.07

- 98** **Challenging**
Make your own timetable.
A ferry leaves the wharf every 40 minutes from 06:50 until 09:30. The trip takes 25 minutes. Write a timetable showing departure and arrival times. Lesson 12.11

Departs	Arrives

YOU DO

On your own: extra working space. Write the question number next to your working.